PULLBACK AND PUSHFORWARD ON SHEAVES OF ALGEBRAIC STRUCTURES IN THE GOOD CASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the sentence 'Adjointness of u_p and u^p is proved in exactly the same way as the proof of Lemma lemma-adjoints-u' concerns one step, whose argument is available verbatim in the cited proof of the same chapter, and the remaining steps are written out in the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The proposition producing a morphism of sites from a functor. Same chapter, Proposition proposition-get-morphism, whose hypotheses the statement assumes. Supplies the morphism of sites and the fact that the colimits defining the left Kan extension are filtered.
- The adjunction on presheaves attached to a functor. Same chapter, Lemma lemma-adjoints-u. Supplies the argument for the adjunction, which is reproduced for algebraic structures.
- Sheafification of presheaves of algebraic structures. Same chapter, the discussion of sheafification of presheaves of algebraic structures preceding the statement. Allows the pullback on sheaves to be defined through sheafification.

Source reference. Original proposition 00X6: sources/stacks-sites/pinned-original-source.tex, source lines 10613–10613. Navigation pointer invoked at proof line 10613; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 00VE: sources/stacks-sites/pinned-original-source.tex, source lines 10632–10632. Navigation pointer invoked at proof line 10632; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Suppose the functor $u : \mathcal{C} \to \mathcal{D}$ satisfies the hypotheses of Proposition 00X6, and hence gives rise to a morphism of sites $f : \mathcal{D} \to \mathcal{C}$. In this case the pullback functor f^{-1} (resp. u_p) and the pushforward functor f_* (resp. u^p) extend to an adjoint pair of functors on the categories of sheaves (resp. presheaves) of algebraic structures. Moreover, these functors commute with taking the underlying sheaf (resp. presheaf) of sets.*

Proof. We have defined $f_* = u^p$ above. In the course of the proof of Proposition 00X6 we saw that all the colimits used to define u_p are filtered under the assumptions of the proposition. Hence we conclude from the definition of a type of algebraic structure that we may define u_p by exactly the same colimits as a functor on presheaves of algebraic structures. Adjointness of u_p and u^p is proved in exactly the same way as the proof of Lemma 00VE. The discussion of sheafification of presheaves of algebraic structures above then implies that we may define $f^{-1}(\mathcal{F}) = (u_p\mathcal{F})^\#$. □